Optimizing Data Integration with an Estimator-Robust Design

Sky Qiu

University of California, Berkeley

November 8, 2024

Joint work with Mark van der Laan, Jens Tarp, Andrew Mertens

Acknowledgement: JICI working group

A-TMLE: van der Laan et al. (2023), https://arxiv.org/abs/2307.12544

RCT + RWD A-TMLE: van der Laan, Qiu, van der Laan (2024), https://arxiv.org/abs/2405.07186

Estimator-Robust Design: in progress

Agenda

1. A quick review of data integration estimators under TMLE framework (FIORD 2022 & 2023)
2. Robustness properties of Adaptive TMLE
3. Hybrid design choices motivated by the robustness properties to optimize data integration
4. Case study: results on augmenting DEVOTE using external real-world data from Optum
5. Discussion

Agenda

1. A quick review of data integration estimators under TMLE framework (FIORD 2022 & 2023)
2. Robustness properties of Adaptive TMLE
3. Hybrid design choices motivated by the robustness properties to optimize data integration
4. Case study: results on augmenting DEVOTE using external real-world data from Optum
5. Discussion

Problem setup

We have access to both an RCT and RWD, each with treatment and control arms. Our goal is to optimize power and precision while rigorously preserving valid inference for the causal estimand.

- $W \in \mathbb{R}^d$, patient baseline characteristics;
- $S \in \{0, 1\}$, study indicator ($S = 1$ for RCT, $S = 0$ for RWD);
- $A \in \{0, 1\}$, binary treatment;
- $Y \in \mathbb{R}$, outcome;
- Observe n i.i.d $O = (W, S, A, Y) \sim P_0 \in \mathcal{M}$.
- We factorize the density of O as follows

$$p(w, s, a, y) = p_W(w) p_S(s \mid w) p_A(a \mid s, w) p_Y(y \mid s, w, a).$$

- We have knowledge on $p_A(a \mid 1, w)$.

Candidate causal estimands

1. Directly pool the RCT and RWD:

$$\tilde{\Psi}(P) = E(Y_1 - Y_0).$$

2. Consider the treatment effect only in the RCT, but pool the covariate distribution W. That is, we average over $p_W = p_{W|S=0} p_{S=0} + p_{W|S=1} p_{S=1}$:

$$\Psi(P) = E_W E(Y_1 - Y_0 \mid S = 1, W).$$

- **Interpretation:** The average treatment effect if everyone in the trial population had been enrolled in the trial.

Comparing their identification assumptions

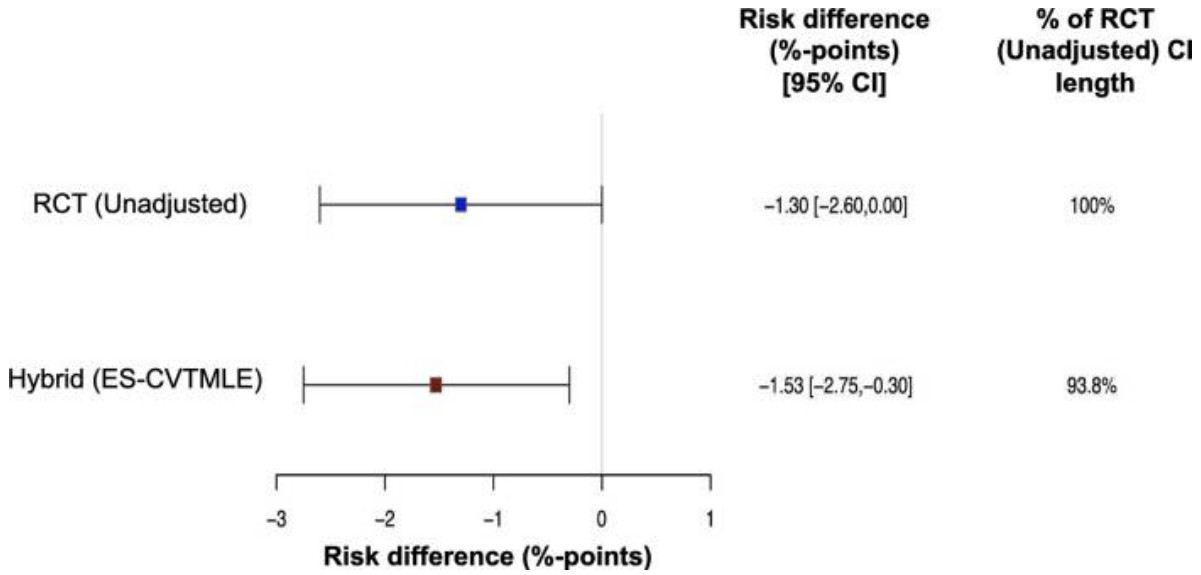
Pooled-ATE	**RCT-ATE**
$\tilde{\Psi}(P) = E(Y_1 - Y_0)$	$\Psi(P) = E_W E(Y_1 - Y_0 \mid S = 1, W)$
Mean exchangeability in the trial (true by design of RCT) $E(Y_a \mid S = 1, W, A = a) = E(Y_a \mid S = 1, W)$	
Positivity of receiving treatment in the trial (true by design of RCT) $0 < P(A = 1 \mid S = 1, W) < 1,\ P_W$-a.e.	
Positivity of trial enrollment $P(S = 1 \mid W) > 0,\ P_W$-a.e.	
Mean exchangeability across the studies $E(Y_a \mid S = 1, W) = E(Y_a \mid W)$	

Challenges: Inconsistencies in outcome measurement between data sources, non-concurrency, unmeasured confounding in the external data, variations in the standard-of-care over time, and other differences between trial and routine care.

Data integration estimators under TMLE framework (1/2)

ES-CVTMLE (Dang et al.) [1], FIORD 2022

- Experiment-Selector Cross-Validated TMLE;
- Deciding whether or not to pool the external data based on a criteria that optimizes bias-variance trade-off. Amount of efficiency gain is driven by the magnitude of the bias.
- Case study [2]: PIONEER 6 (non-inferiority RCT) + Optum to study risk difference of MACE between oral semaglutide and standard-of-care.



Data integration estimators under TMLE framework (2/2)

A-TMLE (M. van der Laan, Qiu, L. van der Laan) [3], FIORD 2023

- Adaptive TMLE was originally proposed by L. van der Laan et al. (2023) as a way to address limited overlap in covariate distributions [4];
- Could be viewed as a regularized TMLE that involves data-adaptively selecting a working model using highly adaptive lasso (HAL) [5].
- Efficiency gain is driven by the complexity of the bias;
- Advantageous in practice: if the bias is small but complex, A-TMLE can still offer large efficiency gains, as even an intercept model may approximate the bias well, maintaining valid inference. Conversely, if the bias is large, methods like ES-CVTMLE tend to reject external data more often, limiting efficiency gains.
- Case study (manuscript in draft):
  - DEVOTE + Optum to study risk difference of MACE between insulin degludec and glargine;
  - Results show that A-TMLE's CI length is **77.2%** of the RCT-only estimate and **79.6%** of the ES-CVTMLE estimate.

A-TMLE for data integration: a quick review

- Target estimand:

$$\Psi(P_0) = E_0[E_0(Y \mid S = 1, W, A = 1) - E_0(Y \mid S = 1, W, A = 0)]$$

- 1st component, **pooled-ATE** estimand:

$$\tilde{\Psi}(P_0) = E_0[E_0(Y \mid W, A = 1) - E_0(Y \mid W, A = 0)]$$

- 2nd component, **bias** estimand:

$$\Psi^{\#}(P_0) \equiv \tilde{\Psi}(P_0) - \Psi(P_0)$$
$$\Psi(P_0) = \tilde{\Psi}(P_0) - \Psi^{\#}(P_0) \text{ bias correction}$$

- We apply A-TMLE to estimate $\tilde{\Psi}$ and $\Psi^{\#}$, respectively.

Agenda

1. A quick review of data integration estimators under TMLE framework (FIORD 2022 & 2023)
2. Robustness properties of Adaptive TMLE
3. Hybrid design choices motivated by the robustness properties to optimize data integration
4. Case study: results on augmenting DEVOTE using external real-world data from Optum
5. Discussion

Robustness properties: projection estimands

It is possible to completely misspecify the working models yet still obtain unbiased estimates!

- For the pooled-ATE estimand $\tilde{\Psi}_{\mathcal{M}_{A,n}}(P_0)$, if $\bar{g}_0(1 \mid W) \equiv P_0(A = 1 \mid W)$ is correctly specified, then the estimator remains unbiased regardless of misspecifications of $\bar{\theta}_0(W) \equiv E_0(Y \mid W)$ or $\tau_{A,0}(W) \equiv E_0(Y \mid W, A = 1) - E_0(Y \mid W, A = 0)$;
- Similarly, for the bias estimand $\Psi^{\#}_{\mathcal{M}_{S,n}}(P_0)$, if $\Pi_0(1 \mid W, A) \equiv P_0(S = 1 \mid W, A)$ is correctly specified, then the estimator remains unbiased regardless of misspecifications of $\theta_0(W, A) = E_0(Y \mid W, A)$ or $\tau_{S,0}(W, A) = E_0(Y \mid S = 1, W, A) - E_0(Y \mid S = 0, W, A)$.

Robustness properties: projection estimands

Key takeaway: if we know the trial enrollment score and the PS in the external data, then we obtain an unbiased estimate.

- To ensure consistency for the projection estimands, we need to know $\bar{g}(1 \mid W)$ and $\Pi(1 \mid W, A)$;
- By Bayes' theorem, $\Pi(1 \mid W, a) =$

$$\frac{g(a \mid 1, W)\bar{\Pi}(1 \mid W)}{g(a \mid 1, W)\bar{\Pi}(1 \mid W) + g(a \mid 0, W)\bar{\Pi}(0 \mid W)},$$

  where $g(a \mid S, W) \equiv P(A = a \mid S, W)$ for $a \in \{0, 1\}$ and $\bar{\Pi}(1 \mid W) \equiv P(S = 1 \mid W)$;
- Since $S = 1$ is the RCT, we know $g(1 \mid 1, W)$ is the trial randomization probability;
- So, to know Π, it suffices to know $\bar{\Pi}(1 \mid W)$ and $g(a \mid 0, W)$;
- If we know $\bar{\Pi}(1 \mid W)$ and $g(a \mid 0, W)$, then we also know $\bar{g}(1 \mid W)$, since

$$\bar{g}(1 \mid W) = g(1 \mid 1, W)\bar{\Pi}(1 \mid W) + g(1 \mid 0, W)\bar{\Pi}(0 \mid W).$$

Robustness properties: nonparametric estimands

- We discussed double robustness w.r.t the projection estimands. But what about w.r.t the true nonparametrically defined estimands?
- It could be shown that when $\Pi(1 \mid W, A)$ is a constant, then we could also claim unbiasedness w.r.t the nonparametric estimands;
- In other words, to obtain an unbiased estimator, we need the trial enrollment score $\Pi(1 \mid W)$ and ps in the external data $g(1 \mid 0, W)$ to be constants;
- **Question: Can we curate a subsample of external patients such that these two quantities are constants within the pooled RCT and curated subsample?**

Agenda

1. A quick review of data integration estimators under TMLE framework (FIORD 2022 & 2023)
2. Robustness properties of Adaptive TMLE
3. Hybrid design choices motivated by the robustness properties to optimize data integration
4. Case study: results on augmenting DEVOTE using external real-world data from Optum
5. Discussion

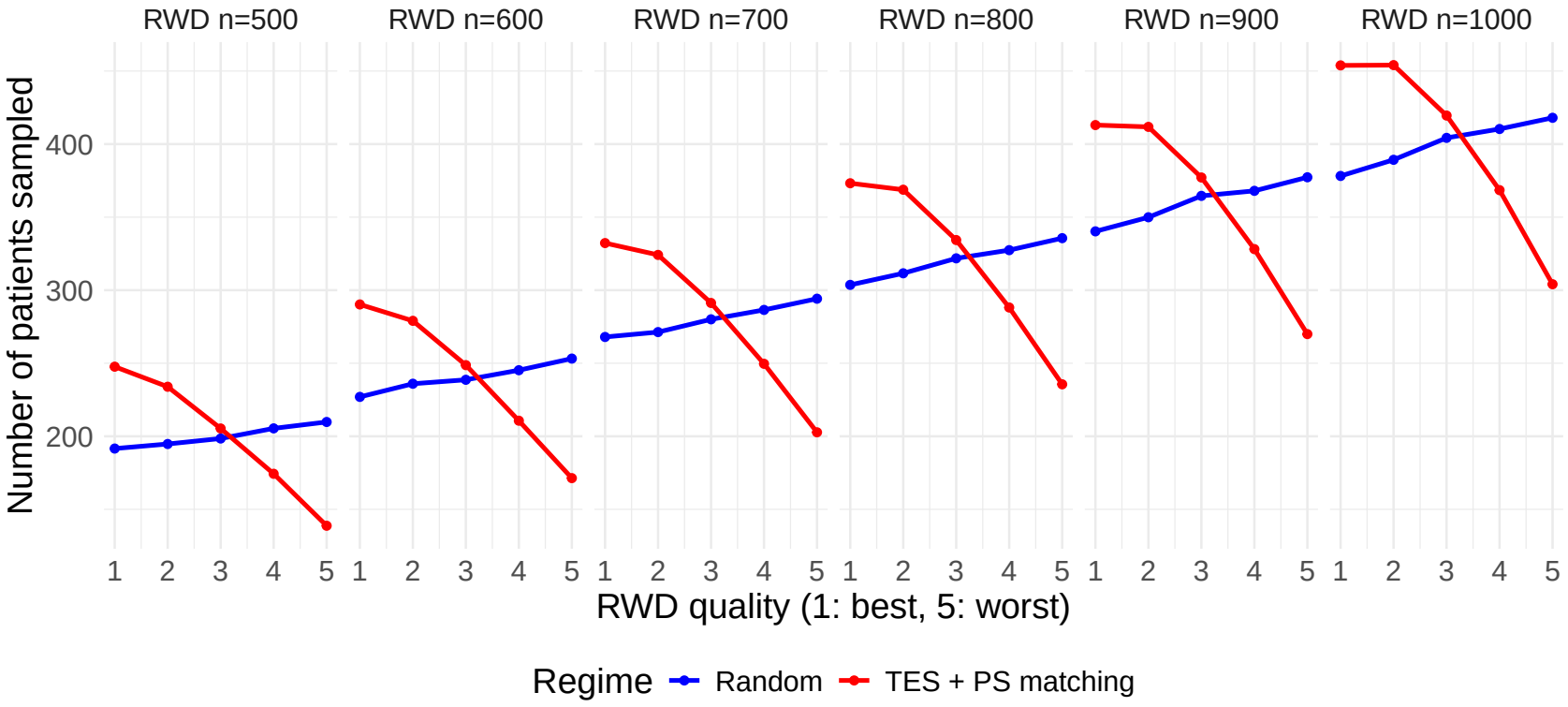
A design inspired by the robustness properties

Robustness analysis of our estimator suggests that it is beneficial to have knowledge on the **trial enrollment score**, $\bar{\Pi}(1 \mid W)$, and the **propensity score in the external data**, $g(1 \mid 0, W)$, or even make them constants. We therefore propose the following design for selecting external patients:

1. Initial filtering of external patients based on inclusion & exclusion criteria;
2. Estimate $\bar{\Pi}(1 \mid W)$ using RCT + filtered external data from 1;
3. Match each RCT patient with $K \gg 1$ external patients based on estimated trial enrollment score;
4. Estimate $g(1 \mid 0, W)$ using RCT + filtered external data from 3;
5. Perform propensity score matching within filtered external data from 4.

Simulation: multiple external data sources

Our proposed strategy samples more individuals from the least biased sources and fewer from the most biased sources.



Number of patients sampled from each external data source using the proposed estimator-robust sampling strategy. External data quality is ranked from 1 (least biased) to 5 (most biased).

Simulation: operating characteristics comparisons

Our proposed strategy recovers the robustness of the estimator in challenging setups.

Sample size of external arms	CI widths			Coverage			Power		
	RCT	Random	Matching	RCT	Random	Matching	RCT	Random	Matching
0	1.169	-	-	0.97	-	-	0.36	-	-
500	-	0.959	**0.725**	-	0.86	**0.95**	-	0.40	**0.85**
600	-	0.923	**0.698**	-	0.85	**0.94**	-	0.35	**0.87**
700	-	0.905	**0.680**	-	0.79	**0.94**	-	0.39	**0.89**
800	-	0.880	**0.679**	-	0.78	**0.95**	-	0.39	**0.89**
900	-	0.881	**0.618**	-	0.84	**0.96**	-	0.42	**0.90**
1000	-	0.859	**0.605**	-	0.80	**0.95**	-	0.38	**0.93**

Candidate designs + analysis methods:

- **RCT:** RCT data + TMLE
- **Random:** randomly sample an equal-sized external treated and control patients + A-TMLE
- **Matching:** trial enrollment score & external propensity score matching + A-TMLE

Agenda

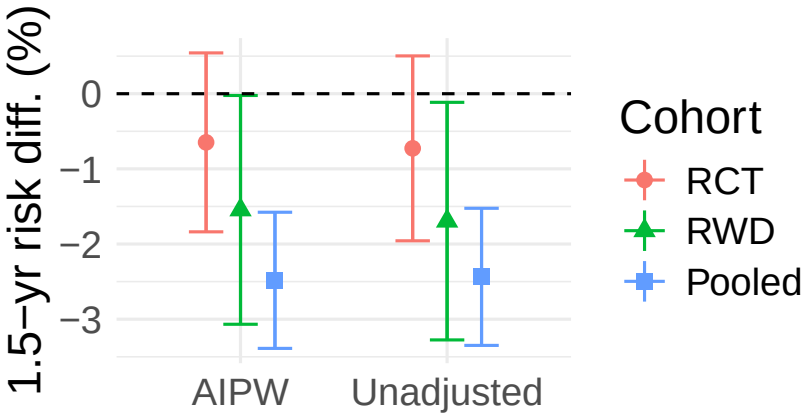
1. A quick review of data integration estimators under TMLE framework (FIORD 2022 & 2023)
2. Robustness properties of Adaptive TMLE
3. Hybrid design choices motivated by the robustness properties to optimize data integration
4. Case study: results on augmenting DEVOTE using external real-world data from Optum
5. Discussion

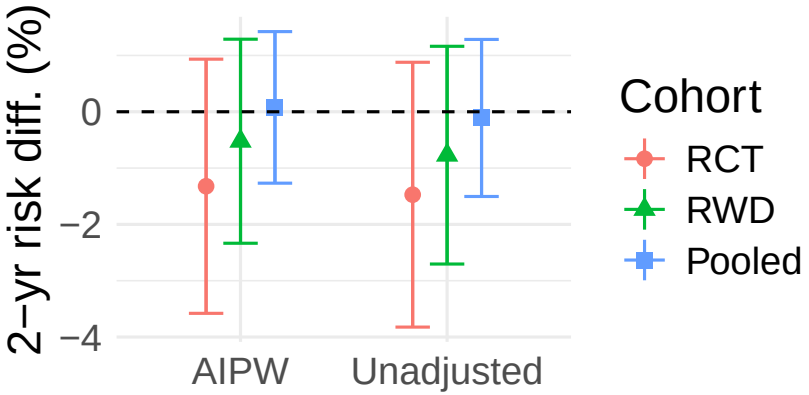
Case study: DEVOTE + Optum

- DEVOTE: evaluate the efficacy and safety of insulin degludec versus glargine in type 2 diabetes [6];
- Treat-to-target, randomized, double-blind, active comparator-controlled, event driven, non-inferiority;
- **Treatment:** insulin degludec; **Comparator:** insulin glargine;
- **Primary outcome:** Composite, first occurrence of an major cardiovascular event (death from cardiovascular causes, nonfatal myocardial infarction, or nonfatal stroke);
- **Scientific question:** 1.5 and 2-year risk differences of MACE between the treatment and comparator arm;
- **External data source:** Optum's de-identified Clinformatics® Data Mart Database, has both degludec treated and glargine treated patients. From 2013 - 2019.

Preliminary analysis

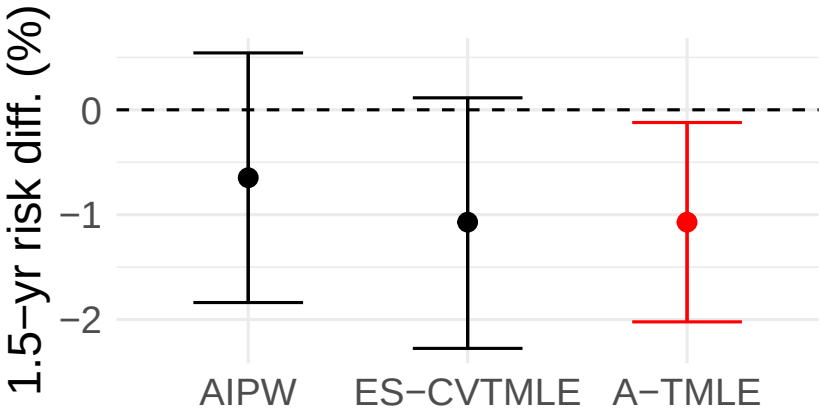
- **External data selection:** 1:3 trial enrollment score matching, 1:5 external treated to control propensity score matching;
- **RCT:** trt. 3818, con. 3817
  RWD: trt. 2359, con. 11795
- The pooled estimates may be biased if there is substantial heterogeneity between the RCT and RWD, violating the identification assumption;
- We expect that to be the case since the overlap in CI between the pooled estimate and RCT estimate is small, especially at 1.5-year.

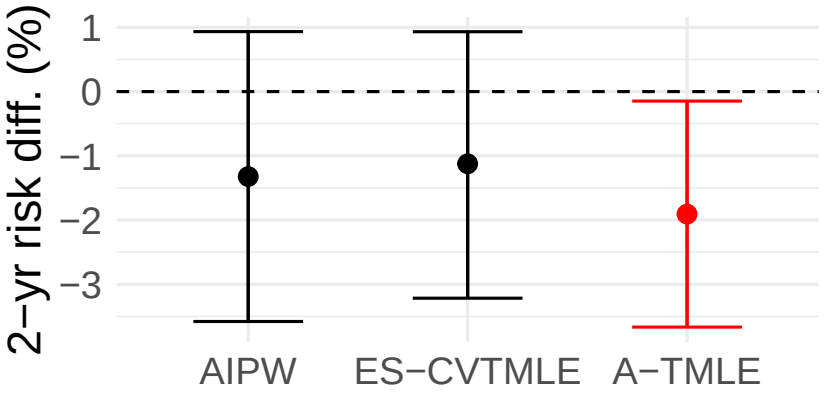




Data integration analysis

- We then compared:
  AIPW: RCT-only data
  ES-CVTMLE: pooled data
  A-TMLE: pooled data
- The results from data integration estimators are aligned with the RCT finding, but with smaller CI widths;
- In particular, A-TMLE showed more than **20%** reduction in CI widths compared to using RCT data alone and showed statistically significant estimate for testing the superiority of insulin degludec in reducing risk of MACE compared to glargine.





Agenda

1. A quick review of data integration estimators under TMLE framework (FIORD 2022 & 2023)
2. Robustness properties of Adaptive TMLE
3. Hybrid design choices motivated by the robustness properties to optimize data integration
4. Case study: results on augmenting DEVOTE using external real-world data from Optum
5. Discussion

All identification assumptions now expected to hold

$$\Psi(P) = E_W E(Y_1 - Y_0 \mid S = 1, W)$$

- Mean exchangeability in the trial (true by design of RCT):

$$E(Y_a \mid S = 1, W, A = a) = E(Y_a \mid S = 1, W), \;\; a \in \{0, 1\}$$

- Positivity of receiving treatment in the trial (true by design of RCT):

$$0 < P(A = 1 \mid S = 1, W) < 1, \;\; P_W\text{-a.e.}$$

- Positivity of trial enrollment (made true by the TES matching step):

$$P(S = 1 \mid W) > 0, \;\; P_W\text{-a.e.}$$

Under the above three assumptions, our target estimand is:

$$\Psi(P_0) = E_0[E_0(Y \mid S = 1, W, A = 1) - E_0(Y \mid S = 1, W, A = 0)].$$

EIC of projection parameter is well-controlled

The P_Π-score component of the canonical gradient of the projection parameter $\Psi^{\#}_{\mathcal{M}_{S,n}}$ at P is given by

$$D^{\#}_{\mathcal{M}_{S,n},\Pi,P} = \frac{A}{\bar{g}_P(1 \mid W)} \tau_{S,\beta(P)}(W,1)(S - \Pi_P(1 \mid W, A)) - \frac{1-A}{\bar{g}_P(0 \mid W)} \tau_{S,\beta(P)}(W,0)(S - \Pi_P(1 \mid W, A)).$$

Note that, under the proposed design,

$$\bar{g}_P(1 \mid W) = \underbrace{g_P(1 \mid 1, W)}_{0.5} \bar{\Pi}(1 \mid W) + \underbrace{g_P(1 \mid 0, W)}_{\approx 0.5,\ \text{PS matching}} \bar{\Pi}(0 \mid W),$$

Therefore, the inversely weighted term $\bar{g}_P(1 \mid W)$ is well-controlled by the external PS matching step in our proposed sampling strategy.

EIC of nonparametric parameter is well-controlled

The efficient influence function of $\Psi : \mathcal{M} \to \mathbb{R}$ at P is given by:

$$D_{\Psi,P}(O) = \bar{Q}_P(1, W, 1) - \bar{Q}_P(1, W, 0) - \Psi(P) + \frac{S}{\bar{\Pi}_P(1 \mid W)} \cdot \frac{2A - 1}{g_P(A \mid 1, W)}(Y - \bar{Q}_P(S, W, A)),$$

where

$$\bar{Q}_P(S, W, A) \equiv E_P(Y \mid S, W, A).$$

Note that the term being inversely weighted is the trial enrollment score and would be well-controlled under our proposed sampling strategy.

Estimator agnostic

Our proposed sampling strategy is not limited to A-TMLE and is likely to benefit any analysis methods one intends to apply. Here is the performance of ES-CVTMLE paired with our proposed sampling strategy:

Sample size of external arms	CI widths			Coverage			Power			Folds pooled (%)	
	RCT	Random	Matching	RCT	Random	Matching	RCT	Random	Matching	Random	Matching
0	1.178	-	-	0.96	-	-	0.44	-	-	-	-
500	-	1.166	**1.067**	-	0.75	**0.95**	-	0.65	**0.58**	35.8	**58.8**
600	-	1.179	**1.068**	-	0.72	**0.95**	-	0.62	**0.60**	39.0	**56.6**
700	-	1.175	**1.081**	-	0.68	**0.95**	-	0.66	**0.60**	37.6	**51.4**
800	-	1.177	**1.084**	-	0.73	**0.95**	-	0.65	**0.60**	35.4	**50.2**
900	-	1.182	**1.084**	-	0.73	**0.95**	-	0.63	**0.62**	36.6	**48.4**
1000	-	1.188	**1.096**	-	0.75	**0.95**	-	0.66	**0.59**	31.2	**46.0**

Table: Average 95% confidence interval widths, coverage, and power by external data sample size and sampling strategy. "RCT" denotes the RCT-only design with the TMLE estimator. "Random" denotes a randomly sampled external cohort of the specified size, analyzed with the ES-CVTMLE estimator. "Matching" denotes the TES + PS matching strategy for sampling the external cohort of the specified size, also analyzed with the ES-CVTMLE estimator.

Summary of advantages

- Ensures the identification assumptions hold, particularly by guaranteeing a non-zero trial enrollment probability in the pooled sample;
- Even with less-than-perfect matching, the EIC for both the projection and original parameters is well-controlled, stabilizing the variance estimate;
- The proposed design choices are flexible and not restricted to a single data integration estimator, meaning various analysis methods can benefit from the design;
- In certain settings, can reduce bias in both magnitude and complexity, as demonstrated by the improvements seen in both ES-CVTMLE and A-TMLE;
- Completely outcome-blind: investigators can try different matching algorithms until the quality of matches is satisfactory.

Conclusion

- Our analysis of A-TMLE's robustness structure shows that, for it to be unbiased with respect to the nonparametrically defined target parameter, both the trial enrollment score $\bar{\Pi}(1 \mid W) \equiv P(S = 1 \mid W)$ and the propensity score in the external data $g(1 \mid 0, W) \equiv P(A = 1 \mid S = 0, W)$ need to be constants;
- This motivates a matching design aimed at making trial enrollment independent of baseline covariates, while ensuring covariates in the external data are balanced within the subsample;
- In simulations, this design restores nominal coverage in difficult scenarios where bias is large, complex, and hard to learn, as well as in cases with low coverage. We observed more than 20% reductions in CI lengths in real-data case study;
- Even with imperfect matching, it controls the terms that are inversely weighted in the EIC, stabilizing variance. Moreover, even if the two terms cannot be made constants but are instead known functions, the estimator remains unbiased with respect to the projection parameter. Numerical evaluations on the next slide show that the oracle bias tends to be small.

Magnitude of the oracle bias: small in practice!

Unobserved			Misspecified			Unobserved + Misspecified		
R^2	Weighted MSE	Oracle bias	R^2	Weighted MSE	Oracle bias	R^2	Weighted MSE	Oracle bias
0.93	9.94×10^{-4}	-6.13×10^{-5}	0.95	1.02×10^{-4}	-9.34×10^{-4}	0.92	1.05×10^{-3}	-1.25×10^{-3}
0.76	3.97×10^{-3}	-1.45×10^{-4}	0.84	4.12×10^{-4}	-1.89×10^{-3}	0.74	4.23×10^{-3}	-2.50×10^{-3}
0.58	8.91×10^{-3}	2.78×10^{-5}	0.72	9.16×10^{-4}	-2.86×10^{-3}	0.56	9.48×10^{-3}	-3.60×10^{-3}
0.44	1.59×10^{-2}	-2.17×10^{-4}	0.61	1.64×10^{-3}	-3.70×10^{-3}	0.42	1.68×10^{-2}	-4.79×10^{-3}
0.33	2.49×10^{-2}	-8.19×10^{-5}	0.54	2.61×10^{-3}	-4.60×10^{-3}	0.31	2.64×10^{-2}	-6.37×10^{-3}
0.26	3.58×10^{-2}	-3.27×10^{-5}	0.48	3.66×10^{-3}	-5.39×10^{-3}	0.24	3.80×10^{-2}	-7.63×10^{-3}
0.20	4.84×10^{-2}	-2.13×10^{-4}	0.44	5.12×10^{-3}	-6.46×10^{-3}	0.19	5.16×10^{-2}	-8.69×10^{-3}
0.16	6.35×10^{-2}	-5.72×10^{-4}	0.41	6.60×10^{-3}	-7.19×10^{-3}	0.15	6.73×10^{-2}	-9.96×10^{-3}
0.13	8.06×10^{-2}	4.09×10^{-4}	0.39	8.38×10^{-3}	-8.35×10^{-3}	0.13	8.54×10^{-2}	-1.12×10^{-2}
0.11	9.91×10^{-2}	9.65×10^{-4}	0.38	1.02×10^{-2}	-9.24×10^{-3}	0.11	1.05×10^{-1}	-1.21×10^{-2}

Table: Numerical evaluations of the oracle bias, i.e., difference between the projection estimand and the nonparametrically defined estimand under various types and degrees of misspecifications of the bias working model.

Bibliography

[1] Dang et al. "A Cross-Validated Targeted Maximum Likelihood Estimator for Data-Adaptive Experiment Selection Applied to the Augmentation of RCT Control Arms with External Data". In: *arXiv.org* (2023). arXiv: 2210.05802 [stat.ME].

[2] Lauren E Dang et al. "Case study of semaglutide and cardiovascular outcomes: An application of the Causal Roadmap to a hybrid design for augmenting an RCT control arm with real-world data". In: *Journal of Clinical and Translational Science* 7.1 (2023), e231.

[3] van der Laan, Qiu, and van der Laan. "Adaptive-TMLE for the Average Treatment Effect based on Randomized Controlled Trial Augmented with Real-World Data". In: *arXiv preprint arXiv:2405.07186* (2024).

Bibliography

[4] van der Laan et al. "Adaptive debiased machine learning using data-driven model selection techniques". In: *arXiv.org* (2023). ISSN: 2331-8422.

[5] Benkeser and van der Laan. "The Highly Adaptive Lasso Estimator". In: *2016 IEEE International Conference on Data Science and Advanced Analytics (DSAA)*. Vol. 2016. IEEE, 2016, pp. 689–696. ISBN: 9781509052066.

[6] Steven P Marso et al. "Efficacy and safety of degludec versus glargine in type 2 diabetes". In: *New England Journal of Medicine* 377.8 (2017), pp. 723–732.